

A Comparative Study of Logistic Regression, Decision Tree, Random Forest and XGBoost for Telecom Customer Churn Prediction

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Abstract—The challenge of predicting whether a customer will churn or not is a real-life binary classification challenge in telecom because it is important that a useful customer churn prediction model not be overwhelmed by the larger non-churn class. In this study, we compare in a simple manner four popular tabular machine learning techniques: Logistic Regression, Decision Tree, Random Forest, and XGBoost, by applying them to the same set of data. This dataset (Telco Customer Churn) is composed of 7,043 customer records and is used for experiments. The identifiers are dropped, the missing values in TotalCharges are replaced by missing values, numeric variables are median imputed and standardized, and categorical variables are mode imputed and one-hot encoded. The train / validation / test partition is fixed as 70/15/15 and all the preprocessing operations are fitted on the train split. Hyperparameters are chosen based on the validation split and the test split is evaluated only once. Random Forest is the best performer with the highest test accuracy (0.7767) and precision (0.5768); Logistic Regression has the highest churn recall (0.7758) and ROC-AUC (0.8439); and XGBoost has the best overall balance, with the highest F1-score (0.6279), macro-F1 (0.7242), balanced accuracy (0.7613), and PR-AUC (0.6644). Random Forest also has a significantly larger train-validation F1 gap (0.2613) compared to the other models. The results indicate that the ranking of the models is strongly dependent on the evaluation goal: when measuring overall accuracy, Random Forest does best while XGBoost or Logistic Regression does best when measuring minority class detection. This is a reminder that for churn prediction, besides accuracy, multiple metrics that are aware of the imbalance are needed.

Index Terms—customer churn, telecommunications, logistic regression, decision tree, random forest, XGBoost, class imbalance, machine learning

I. INTRODUCTION

Loss of customers from a services provider is referred to as customer churn and has been well studied as a predictive and decision support problem in the telecommunication industry [1]–[3]. While providing the providers with an estimated churn risk, so that they prioritize retention activity, churn prediction is operationally valuable doesn't mean that a retention campaign will be profitable or effective [4]–[6]. Therefore, churn modelling shouldn't be seen as a competition on accuracy, but rather as a supervised learning problem that is part of a bigger problem in managing customers.

There is a vast amount of literature on predicting churn using statistical models, decision trees, ensembles, support-vector machines, neural networks and hybrid feature-selection

or resampling approaches [7]–[14]. The tree ensembles and other tabular machine-learning approaches have been reported to be very effective in recent studies, along with the need for explainability and reproducible evaluation [15]–[18]. It is however challenging to compare directly across published papers as datasets, preprocessing, sampling strategies, decision thresholds, train/test splits and reported metrics vary.

Logistic Regression, one single Decision Tree, Random Forest and XGBoost are thus the models that are compared with this paper focusing on a deliberately narrow question: *how do four common classical machine-learning algorithms behave when trained and evaluated under the same leakage-safe pipeline on the same churn dataset?* A wide range of models exists from the linear model, a probabilistic model, to a simple interpretable tree, a bagged tree ensemble, and a gradient boosted tree ensemble (1.94,24,25,26,27)

The main contributions are threefold. The first is that they share the same data partition, stratified by model, and a pipeline of pre-processing tasks was trained for them, minimizing unnecessary experimental confounding. Second, with the imbalanced target, accuracy and precision for churn are estimated, along with precision and recall for churn (PR-AUC, ROC-AUC, confusion matrices) to evaluate the performance of the model [24]–[27]. Third, validation behaviour and train-validation gaps are reported to account for generalization, not just test accuracy, when choosing a model.

II. RELATED WORK

A. Churn prediction in telecommunications

Prior churn research has shown that the predictive approach, the set of attributes that can be used, and the nature of the evaluation goal can significantly affect the set of individuals considered high risk [3]. Ahn *et al.* discussed the issue of churn determinants in mobile telecommunications and Hadden *et al.* explored computer assisted churn management and emphasized on the importance of churn management in the retention system with predictive modeling [1], [2]. Lemmens and Croux demonstrated that the bagging and boosting of classification trees could help enhance churn prediction power and they explained balanced sampling for rare churn events [7].

The problem of class imbalance has continually plagued. Burez and Van den Poel showed that it is not sufficient to

use the raw accuracy for evaluating churns and investigated sampling, gradient boosting, and weighted forests [8]. Xie *et al.* proposed improved balanced random forests, and Idris *et al.* used the PSO algorithm as a balancing and feature selection method to integrate balanced random forest [9], [11]. Other comparisons looked at various classifiers and feature representations over the telecom data, with similar results: conclusions will vary upon the data and experimental protocol used [10], [12], [13].

This theme has been followed in more recent studies. Ahmad *et al.* introduced a big-data platform for their machine-learning churn framework [14]. Wagh *et al.*, Chang *et al.*, and Poudel *et al.*, have explored classification techniques for telecom churn, tree and ensemble models, and tabular prediction and explainability, respectively [15]–[17]. Barsotti *et al.* recently surveyed a decade of TelCo churn-prediction techniques, and the variety of modeling approaches in use in the field [18].

B. Algorithms considered in this study

Logistic Regression is a model of the log-odds of a binary response as a linear function of the predictors, and a good baseline model if the predictors are one-hot encoded and include informative data [19], [28]. Decision Trees recursively split the feature space and can be expressed in an interpretable way [20] for the purpose of nonlinear interaction. To combat the variance of the individual trees, Random Forest combines predictions from a number of trees trained on random bootstraps of the data, as proposed by [21]. Gradient boosting, on the other hand, constructs a sequence of weak learners to minimize a specific loss [22] and XGBoost is an efficient and regularized implementation with system-level optimizations for scalable tree boosting [23].

Such models are ideal for customer data that is well-organized. They are also complementary in their ideas of feature importance: Logistic Regression coefficients represent signed linear relationships from encoded and scaled features, whereas tree models show importance based on splits. These measures must be read as internal relationships within the model, and not as causality. Further post hoc methods, such as LIME and SHAP, exist if more detailed explanations of the local or more coherent feature attributions are needed [29], [30].

C. Evaluation under class imbalance

It is possible to achieve high accuracy when the outcome is imbalanced, and low accuracy when the minority class is detected with low accuracy. Imbalanced learning reviews have thus pleaded for class sensitive metrics and training, which would reveal performance on both classes [24], [31]. While ROC analysis gives an overview of ranking performance over thresholds, Precision-Recall analysis can provide more useful information when the positive class is relatively rare as it directly addresses positive predictive value and sensitivity [26], [27]. In this context, the present study also presents the PR-AUC and threshold-based metrics and raw confusion counts in addition to the ROC-AUC.

TABLE I
STRATIFIED DATA PARTITION

Split	Customers	Churn	Churn rate
Train	4,930	1,308	26.53%
Validation	1,056	280	26.52%
Test	1,057	281	26.58%

III. MATERIALS AND METHODS

A. Dataset

The experiment is done on public Telco Customer Churn data distributed for analytics demos by IBM [32]. The local copy has 7,043 rows and 21 columns, one of which (called *Churn*) is a binary variable and another one is a unique customer identifier. Of the 7,043 customers, 5,174 (73.46%) are non-churners and 1,869 (26.54%) are churners. Therefore, if the majority class only classifier is used, the accuracy will be about 73.46

The identifier *customerID* is eliminated as it serves only to identify who the customer is, and is not a re-usable behavioral predictor. The field *TotalCharges* is loaded as a text field due to missing values, and then converted to numeric with the invalid strings being mapped to missing values. This leaves 11 missing *TotalCharges* values. There were no duplicate rows or customer identifiers in the input data.

B. Data partitioning and leakage control

Fixed random seed of 42 used. The data is divided using stratified sampling procedures into 70

All the imputation, scaling and categorical encoding are fitted only on the training data. The fitted transformer is then applied to the validation and test partitions without any changes. With this, information from the held-out sets will not affect the preprocessing parameters.

C. Preprocessing

The numeric variables are: *SeniorCitizen*, *tenure*, *MonthlyCharges* and *TotalCharges*. Numeric values that are missing are filled in with the median of the training-set and numeric columns are transformed into zero-centered, unit-scale values. There are 15 other predictors that are categorical. The missing values of the categorical variables are filled with the most common training value and the categorical variables are one-hot encoded; categories that do not occur in the training data are not written to the models. The final transformed input has 45 model features.

Data handling and pre-processing, model fitting and evaluation are implemented by *pandas/NumPy*, *scikit-learn*, *XGBoost* is used as the boosted-tree model, and *Matplotlib* is used for visualization [23], [35]–[38].

D. Model configuration and selection

Table II summarizes the model-selection procedure. The value of the inverse regularization parameter *C* for Logistic Regression is chosen from the set $\{0.001, 0.01, 0.1, 1, 10, 100\}$ by validation F1. *C* = 0.1 is the value selected. The number of iterations is 3000 and a balanced class weight is used.

The Decision Tree is trained with a maximum depth between 1 and 15, a minimum leaf size of 5 samples, and an equal weight of the classes, bottom depth 3 has the highest validation F1. Random Forest is tested for 25, 50, 100, 200, 300 and 500 trees with a minimum leaf size of 2 with balanced class weights; 50 trees are chosen. In the case of XGBoost, the ratio of classes in the training set is 2,7691 which corresponds to the parameter `scale_pos_weight`. The number of boosting rounds is set to a maximum of 1000 with a learning rate of 0.03, a maximum depth of 4, a minimum child weight of 2, a row subsampling value of 0.85, a column subsampling value of 0.85, an L_1 regularization value of 0.05, and an L_2 regularization value of 1. With patience=50, the model is early stopped at iteration 213, based on the validation log-loss.

E. Evaluation metrics

Let TP represent the number of true positives, FP represent the number of false positives, TN represent the number of true negatives, and FN represent the number of false negatives. The precision, recall and F1 values for churn are:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad (1)$$

$$\text{Recall} = \frac{TP}{TP + FN}, \quad (2)$$

$$F_1 = 2 \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (3)$$

Balanced accuracy is the average of the sensitivity and specificity. Macro-F1 treats the two classes equally, while ROC-AUC and PR-AUC measure the ranking behavior based on threshold-independence from different perspectives [25]–[27]. The prevalence of churn is low at 26.54%, so there is no single metric used for churn.

IV. RESULTS

A. Validation behavior and model selection

The model-selection curves for each of the models are shown in Figure 1. The best validation F1 value of Logistic Regression is $C = 0.1$ while the best value for shallow Decision Tree is depth 3. The performance of Random Forest is best at 50 trees and increases with the number of trees afterwards but the train-validation difference seems to be increasing with the number of trees. The validation log-loss of XGBoost continues to decrease up to the early stopping point around iteration 213.

The finally selected models have a validation F1 score of 0.6364 for XGBoost, 0.6247 for Logistic Regression, 0.6218 for Decision Tree, and 0.6208 for Random Forest. The small gap in the validation F1 score suggests that it is necessary to look at the results on the test set that were not used in the training set with several criteria before choosing a model based on the accuracy score.

B. Final test performance

The results on a locked test set are reported in Table III. There are no models that excel on all measures. Random Forest achieves the highest accuracy (0.7767), and churn

precision (0.5768), but its churn recall is low (0.6014). Logistic Regression has the best churn recall score (0.7758) and ROC AUC score (0.8439). XGBoost gives the highest churn F1 (0.6279), macro-F1 (0.7242), balanced accuracy (0.7613), and PR-AUC (0.6644), while maintaining recall of 0.7687. Decision Tree is competitive in recall but is lower than the other models on ROC-AUC and PR-AUC.

The numerical difference also shows the importance of having an objective of deployment before selecting the classifier. Logistic Regression and XGBoost are better at the default threshold if false negatives are deemed to be more expensive as they don't reach back to the churner. Random Forest's ability to reduce false alarms might be desirable if a retention team has limited capacity and prefers to have fewer false alarms, but may be offset by threshold tuning.

C. Confusion-matrix analysis

All four confusion matrices are presented in the same row in Figure 2 for easy comparison. Logistic Regression correctly identifies 218 of 281 churners and fails to identify 63 churners, while giving 200 false positives. XGBoost identifies 216 churners and misses 65 and misidentifies 191 churners. Decision Tree is able to identify 208 churners while having 73 false negative results. Random Forest has a much lower churn recall at the expense of churn accuracy and precision due to its 112 churners missed and 124 false positive churn predictions.

D. ROC and Precision-Recall behavior

The ROC and Precision-Recall curves are shown in Fig. 3. Logistic Regression has the best ROC-AUC of 0.8439, followed by XGBoost at 0.8376, Random Forest at 0.8286 and Decision Tree at 0.8085. On PR-AUC, which is very useful for minority churn class (0.6644) [27] is ranked first, followed by Logistic Regression (0.6566) and Random Forest (0.6367). The distance to Decision Tree trails is: 0.5474.

E. Generalization and feature importance

The train-validation F1 gap is 0.0027 for Logistic Regression, 0.0012 for Decision Tree, 0.2613 for Random Forest and 0.0456 for XGBoost. This Random Forest gap is then a lot bigger than for the other models, which is consistent with a model that fits the training data very well, but is not transferred as well to held-out data. The chosen shallow Decision Tree is the most regularized model with virtually no gap, but with slightly less test discrimination power.

A number of variables repeat themselves in the model-specific importance analyses. Shorter term correlates with changes in Logistic Regression coefficients, and month-to-month contracts and fiber-optic Internet are also given significant model weight, as are electronic-check payment and not having online-security or technical-support services. Tenure, total charges, monthly charges, month to month contract status and online-security status are some of the most important variables for Random Forest. Month-to-month contracts are XGBoost's most impactful split, followed by fiber-optic service

TABLE II
MODEL CONFIGURATION AND VALIDATION-BASED SELECTION

Model	Candidate / fixed settings	Selection criterion	Selected configuration
Logistic Regression	$C \in \{0.001, 0.01, 0.1, 1, 10, 100\}$; balanced class weight; max iter. 3000	Validation F1	$C = 0.1$
Decision Tree	max depth 1–15; min samples/leaf 5; balanced class weight	Validation F1	max depth = 3
Random Forest	trees $\in \{25, 50, 100, 200, 300, 500\}$; min samples/leaf 2; balanced class weight	Validation F1	50 trees
XGBoost	1000 max rounds; LR 0.03; depth 4; subsample 0.85; colsample 0.85; class weight ratio 2.7691	Validation log-loss with early stopping	best iteration = 213

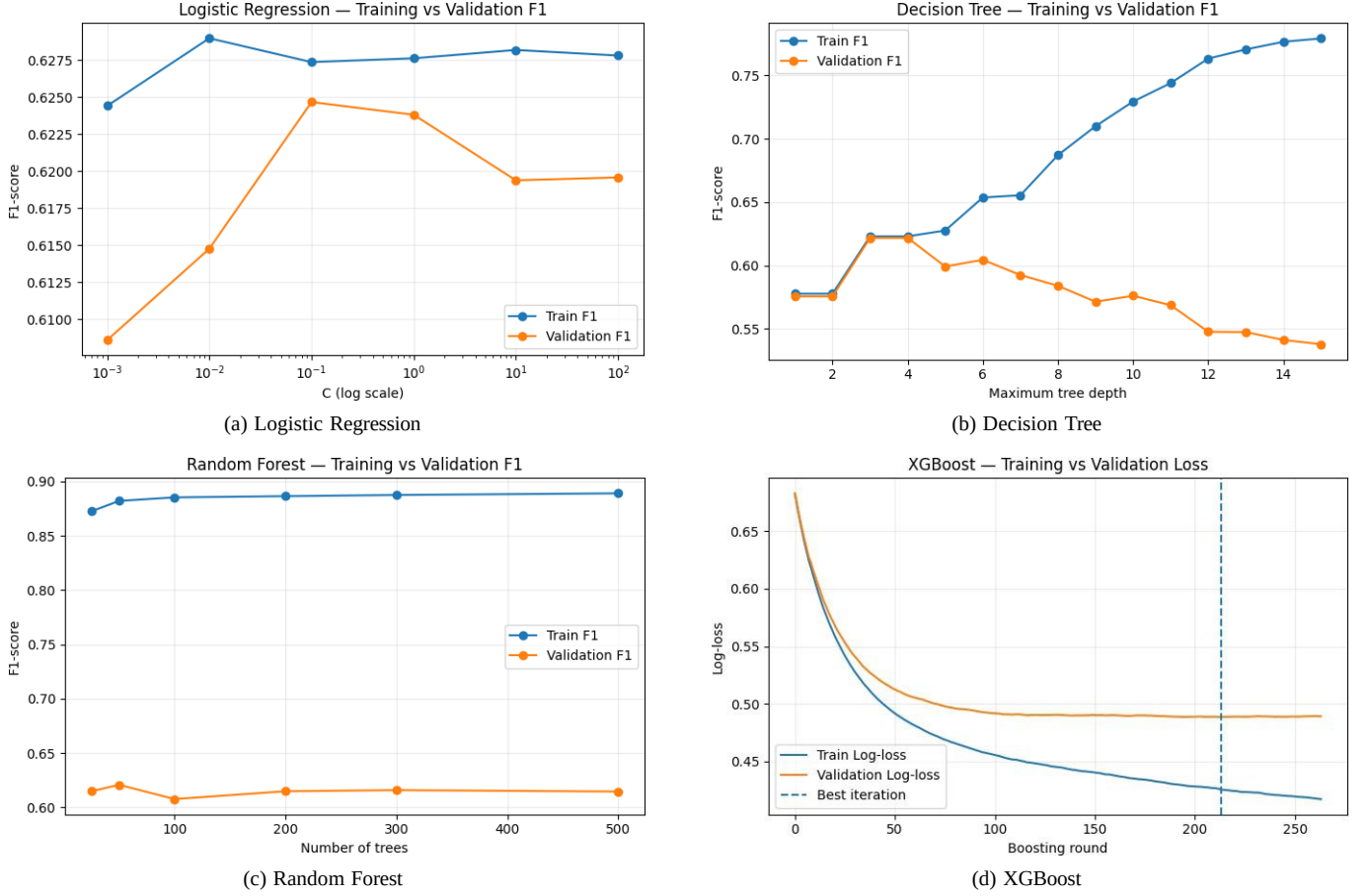


Fig. 1. Model-selection behavior. The first three panels show training and validation F1 across the tuned hyperparameter. XGBoost shows training and validation log-loss across boosting iterations with early stopping.

and lack of online security and technical support. The model-specific rankings are provided as supplemental plot files in the source package included with this report.

V. DISCUSSION

The main finding is that there is no single algorithm that is always better, but there are metrics that support one model over another. The accuracy of Random Forest is the highest is 0.7767 but it correctly identifies just 60.14 of churners. This is because the test set consists of 776 non-churners and 281 churners and a model can improve the aggregate accuracy by correctly classifying the majority class while compromising the accuracy

for the minority class (churners). In the churn literature, the focus is on class-imbalance methods and imbalance-aware evaluation, both of which have similar motivations as the one given above.

At the default decision threshold, XGBoost has the best balance. It outperforms Logistic Regression on both F1 and macro-F1, balanced accuracy and PR-AUC, and is near the Logistic Regression on churn recall. This is consistent with its relatively good performance of learning nonlinear interactions and correcting the residual error in sequence [22], [23]. However, the difference between Logistic Regression and it is not significant. The highest ROC-AUC and Churn recall and

TABLE III
FINAL TEST-SET PERFORMANCE

Model	Accuracy	Precision	Recall	F1	Macro-F1	Bal. Acc.	ROC-AUC	PR-AUC
Logistic Regression	0.7512	0.5215	0.7758	0.6237	0.7189	0.7590	0.8439	0.6566
Decision Tree	0.7474	0.5174	0.7402	0.6091	0.7112	0.7451	0.8085	0.5474
Random Forest	0.7767	0.5768	0.6014	0.5889	0.7178	0.7208	0.8286	0.6367
XGBoost	0.7578	0.5307	0.7687	0.6279	0.7242	0.7613	0.8376	0.6644

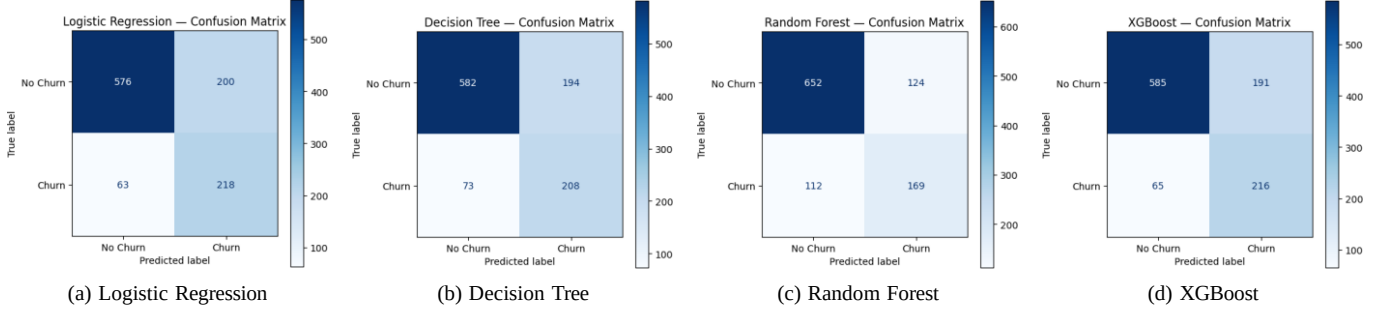


Fig. 2. Test-set confusion matrices. Rows are actual classes and columns are predicted classes. All models are evaluated on the same 1,057-customer held-out test split.

also very small train-validation F1 gap are achieved when the model is Logistic Regression. Logistic Regression is a viable choice for an organization that has a commitment to a baseline that is transparent, stable and sensitive to churners.

It is worth noting that the generalization gap of the Random Forest is quite large. Its training F1 of 0.8821 versus validation F1 of 0.6208 produces a gap of 0.2613, far larger than the gaps for XGBoost (0.0456), Logistic Regression (0.0027), and Decision Tree (0.0012). This training behavior suggests that whilst Random Forest has the best test accuracy this shouldn't be read as a blanket superiority in generalization. Additional regularization and threshold optimization, resampling and probability calibration would be sensible next experiments.

The outcome has a business connotation as well. A risk score does not necessarily represent a policy of retention. Past studies have found that maximizing the profitability of churn interventions should account for the cost of the campaign, as well as its profitability, [4] and field experiments have found that customers who are most at risk are not necessarily the ones who respond to a retention campaign most effectively [5], [6]. The present study thus validates the use of the classifier as a ranking/screening tool prior to business-rule, uplift or treatment-effect analysis for direct intervention.

Last, the recurring feature patterns offer operational hypotheses and not causal conclusions. Several of the fitted models give significant importance to month-to-month contracts, tenure, service configuration, and payment method. These associations can guide analysts in plotting the focus of their search for indicators of customer experience, but the importance of these features depends on how the features are encoded, whether they are correlated with each other, the model structure, and the sample composition. The use of model agnostic explanation methods like LIME or SHAP could enhance future analysis [29], [30].

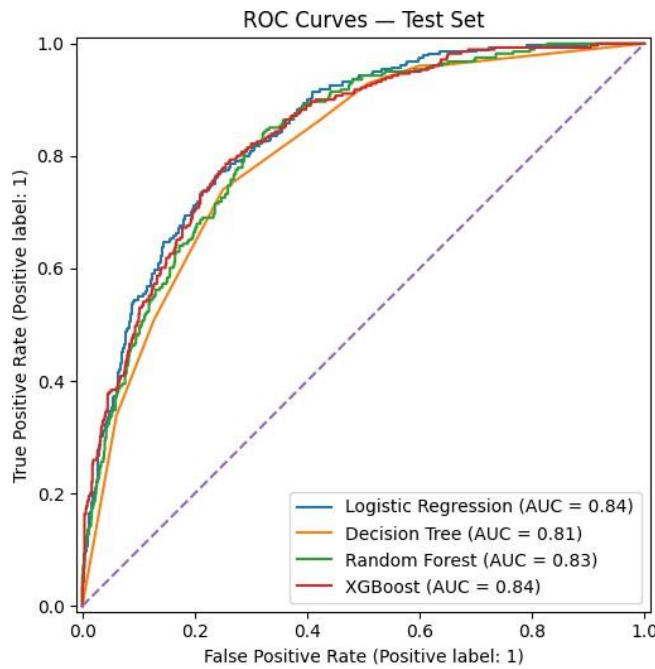
VI. LIMITATIONS

There were several restrictions to the conclusions. The first one is not a real-time telecom cohort, and the sample provided by IBM is for demonstration purposes with analytics [32]. It is thus not known whether it is generalizable to another operator, country, tariff system or time period. Second, the experiment employs one fixed stratified split and one random seed; repeated cross validation or repeated holdout would yield uncertainty estimates and more accurate model ranking [33]. Third, the hyperparameter searches are only partial to keep them reasonably small and not exhaustive, meaning that better hyperparameters may be found for each algorithm.

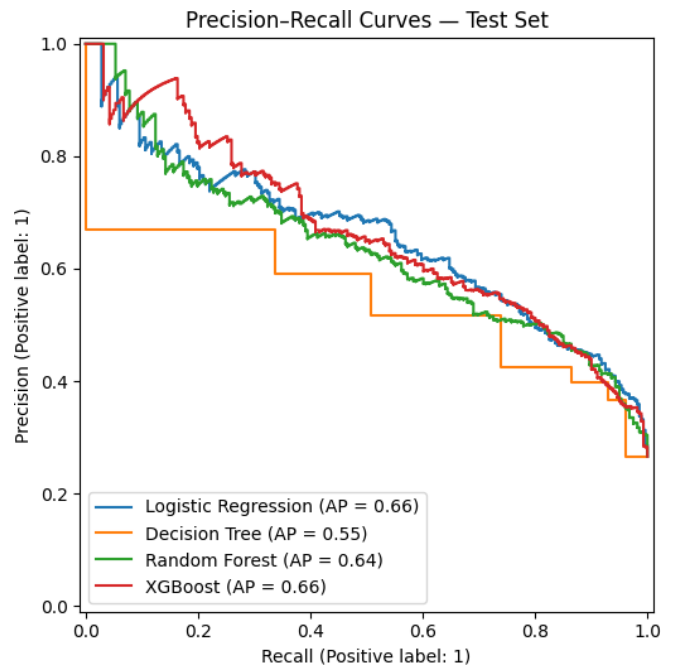
Fourth, all the results based on thresholds are calculated with the default classification threshold. The threshold should be based on the relative costs of a false positive retention offer and a false negative missed churner in practice. Fifth, it is not directly compared with a variety of imbalance strategies including SMOTE, undersampling or cost-sensitive optimization [24], [31] but rather class weighting is implemented. Lastly, feature importance plots offer a summary of the model but do not offer formal causal or counterfactual explanation.

VII. CONCLUSION

A comparison was conducted between four popular machine learning models—Logistic Regression, Decision Tree, Random Forest, and XGBoost—to determine which model performed best for the telecom customer churn problem, all with the same leakage-safe preprocessing, splitting, and evaluation procedure. The highest accuracy and precision is obtained by the Random Forest algorithm on the test set with held-out data, while for the highest churn recall and ROC-AUC, the algorithm of Logistic Regression is used, and for the highest F1, macro-F1, balanced accuracy, and PR-AUC, the algorithm of XGBoost is used.



(a) Receiver-operating-characteristic curves



(b) Precision-Recall curves

Fig. 3. Threshold-independent test-set discrimination. ROC and Precision-Recall views lead to slightly different model rankings because the positive class is imbalanced.

Therefore, the preferred model is one that is defined by the operational goal and not just a single headline number.

In a setting where the detection of minority class is the key step, XGBoost offers the best overall balance in this experiment, while Logistic Regression is still a viable and reliable option. Random Forest has also the lower churn recall and the higher train-validation gap, which should be taken into consideration with the high accuracy. Further studies are needed before deployment that examine repeated splits, calibrated probabilities, industry-specific decision thresholds, more comprehensive explanations, and external telecom data sets.

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